Big Issues in Quantum Algorithms and Quantum Error Correction

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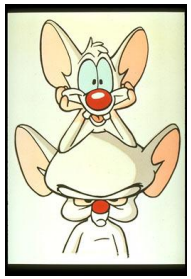




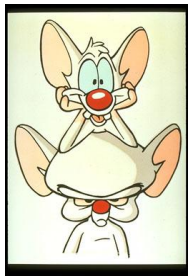
Quantum Algorithms

“If I had a good idea in quantum algorithms, that is what I’d be working on” – John Preskill 2004





Quantum Algorithms



1. The Question is easy: where are all the quantum algorithms?

There has been no second quantum algorithms revolution. Why? What should we do to fix this?

Scott Aaronson (Waterloo)
Dorit Aharonov (Hebrew U)
Andris Ambainis (Waterloo)
Dave Bacon (Washington)
Bob Beals (Princeton)
Charlie Bennett (IBM)
André Berthiaume (DePaul)
Ethan Bernstein
Thomas Beth (Karlsruhe)
Eli Biham (Technion)
Ofer Biham (Hebrew U)
Mark Byrd (SIU)
David Biron (Hebrew U)
Michel Boyer (Montreal)
Gilles Brassard (Montreal)
Samuel Braunstein (York)
Harry Buhrman (CWI)
Nicolas Cerf (Brussels)
Andrew Childs (Caltech)
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Don Coppersmith (IBM)
Claude Crépeau (McGill)
Wim van Dam (Santa Barbara)
Christopher Dawson (UQ)
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Veit Elser (Cornell)
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Umesh Vazirani (UC Berkeley)
Haobin Wang (UC Berkeley)
John Watrous (Calgary)
Avi Wigderson (IAS)
Lianao Wu (Toronto)
Ronald de Wolf (CWI)
Andrew Yao (Tsinghua)
Christof Zalka (Waterloo)

Where Are the Quantum Algorithms?

Early days, TWO tools:

quantum Fourier transform (used in Shor's algorithm)

amplitude amplification (used in Grover's algorithm)

New tools have been developed:

quantum random walks

adiabatic algorithms

"Tools" or "tricks" for quantum algorithms are still limited.

More work on primitives, circuit constructions

Example: the use of quantum random walks lead to algorithms for element distinctness, finding triangle in graphs, matrix multiplication verification…

[Decker, Janzing, Roetteler 04]

Efficient circuits for symmetric POVMs

[Bacon, Harrow, Chuang 05]

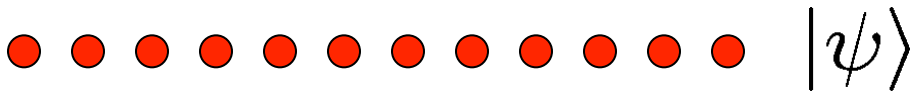
Efficient circuits for the Schur transform

Where Are the Quantum Algorithms?

B. (Pure state) entanglement is necessary (but not sufficient) for quantum speedups.

Better language for entanglement (and interference)

We are missing an effective machine language



Lesson: multipartite entanglement classification hard.

Quantum data structures [Aaronson 03]

Note: In current quantum algorithms, the oracle call hides the role entanglement is playing

Where Are the Quantum Algorithms?

C. Where is the boundary between classical efficiently simulatable and quantum speedup?

In addition to learning about the quantum speedup

1. Teaches us why our classical algorithms succeed and fail
at simulating quantum systems.
Better classical algorithms.
[Vidal 03, Osborne 05] Efficient simulation of one-dimensional quantum many-body systems

2. Teaches us what is and what is not a quantum computer.
Practical consequences

Impact of Quantum Algorithms

Classical computational complexity problems approached from a quantum perspective.

[Kerenidis,de Wolf 2003], [de Wolf, Wehner 2005]
bounds for locally decodable codes, private information retrieval

[Aaronson 2004]
simple proof of closure of PP under intersection

[I. Kerenidis, 2005]
lower classical circuit bounds from quantum communication complexity

[S. Laplante, T. Lee, and M. Szegedy, 2005]
formula size lower bounds from quantum lower bounds

[van Dam 2004]
quantum circuits and the Reiman hypothesis

Quantum complexity is to classical complexity as complex analysis is to real analysis?

Future Quantum Algorithms?

Sampling of open problems (mostly from Scott Aaronson's list: http://www.scottaaronson.com/writings/qchallenge.html)

1. PAC learnability of AC0 or TC0 circuits?
2. Graph isomorphism
3. Linear code isomorphism
4. Certain shortest vector in a lattice problems
5. Better than Grover speedups for total functions?
6. Questions in classes like PPP, PPA, PPAD, PLS?
   (finding Nash equilbria, market eqiulibria, …)

What advantages do the polynomial speedups offer?

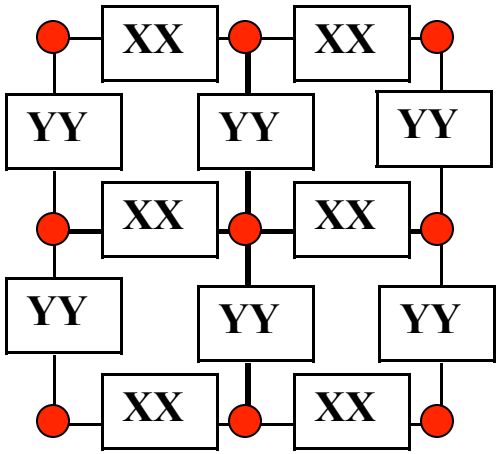
A quantum 1Hz computer vs a classical 10 GHz computer?

Quantum Error Correction

Heretic: the fundamental question in quantum error correction is <u>not</u> what the threshold is.

The fundamental question is either
(1) what is the proper architecture for quantum error correction?
or
(2) What is the proper physics for quantum error correction?

Example: Quantum error correction is essentially cooling.



$$H = \sum_{ij} X_{i,j}X_{i+1,j} + Y_{i,j}Y_{i,j+1}$$

Quantum error correcting subsystem.

If cooling faster than decoherence, then this is equivalent to a single qubit quantum error correcting code.